

Gradient Descent's Second Iteration

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1 Compute $\hat{y}$

New $w = 0.182$

New $b = 0.0645$

$$\hat{y}_i = wx_i + b$$

$$\hat{y}_1 = 0.182 \cdot 1 + 0.0645 = 0.2465$$

$$\hat{y}_2 = 0.182 \cdot 2 + 0.0645 = 0.4285$$

$$\hat{y}_3 = 0.182 \cdot 3 + 0.0645 = 0.6105$$

$$\hat{y}_4 = 0.182 \cdot 4 + 0.0645 = 0.7925$$

2 Gradient Descent

$$\frac{\partial L}{\partial w} = \frac{2}{n} \sum_{i=1}^n (wx_i + b - y_i) \cdot x_i$$

$$\frac{\partial L}{\partial b} = \frac{2}{n} \sum_{i=1}^n (wx_i + b - y_i)$$

$$\begin{aligned} \frac{\partial L}{\partial w} = \frac{2}{4} & \left[(0.182 \cdot 1 + 0.0645 - 2) \cdot 1 + (0.182 \cdot 2 + 0.0645 - 2.8) \cdot 2 \right. \\ & \left. + (0.182 \cdot 3 + 0.0645 - 3.6) \cdot 3 + (0.182 \cdot 4 + 0.0645 - 4.5) \cdot 4 \right] = -15.1475 \end{aligned}$$

$$\begin{aligned} \frac{\partial L}{\partial b} = \frac{2}{4} & \left[(0.182 \cdot 1 + 0.0645 - 2) + (0.182 \cdot 2 + 0.0645 - 2.8) \right. \\ & \left. + (0.182 \cdot 3 + 0.0645 - 3.6) + (0.182 \cdot 4 + 0.0645 - 4.5) \right] = -5.411 \end{aligned}$$

3 Compute new w and b

$$w = w - \alpha \cdot \frac{\partial L}{\partial w}$$

$$w = 0.182 - 0.01 \cdot -15.1475 = 0.333475$$

$$b = b - \alpha \cdot \frac{\partial L}{\partial b}$$

$$b = 0.0645 - 0.01 \cdot -5.411 = 0.11861$$